

X'mas SnowGlobe TableTop Toy for Raspberry Pi Pico

for Pico/W Pico2/W

- *User Manual* -

(1 ~ 2 Hours)



<https://.../Raspberry Pico Edition>



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Maintained By: Percy Chen

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Santa & His Elf

- Part 1 -

Introduction



Why SnowGlobe?



Because every learner needs a moment of wonder. A spark that turns curiosity into the courage to create🔨.

This **Product** was born from a simple observation: when students build something that feels special ✨, they **stay engaged** longer, **ask deeper questions**, and **take greater pride** in their **learning**.

A snow globe captures that feeling **perfectly**👍:

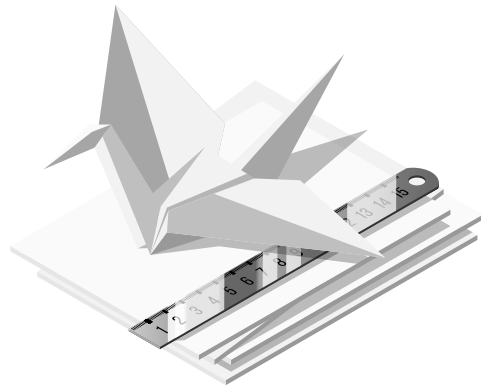
Shake it, and a quiet world🌍 comes alive. **Build one**, and you create your own tiny universe🪐 of possibilities.

X'mas SnowGlobe TableTop Toy takes that magic ✨ and turns it into a hands-on STEM story. Students aren't just assembling parts; they're ultimately crafting a piece of art powered by their own code - They see how a line of logic becomes a combination of colorful light, how sensors respond to their touch or environment, and how learning STEM (Coding) can transform something **DULL** to something **SPECTACULAR**.

The SnowGlobe becomes a **symbol**⭐ of what they can build when **imagination** meets **engineering**.

SnowGlobe isn't just a project! It's an invitation to create something meaningful, memorable, and a little bit magical. Treat it in the end as a keepsake - a reminder of a moment when you, a learner, discovered **"I made this myself."**🎄





Pre-requisite for the Guided-tour

This project might be unlike anything you have done before. It involves **two** disciplines from the acronym **STEAM**—**T**echnology & **A**rt. Thus in our opinion, it would benefit greatly if you approach this project challenge armed with some prior experience & knowledge.

1. The construction & assembly portion will be a breeze if you have taken apart your toys or done simple electronics in the past. However, if you want to brush up your skill in doing electronics surgery or handling electronics, you can refer to the following link:

Dissect An Old Toy Series by Scitech WA:

[Dissect An Old Toy Part 1 : Think](#)

[Dissect An Old Toy Part 2 : Build](#)

[Dissect An Old Toy Part 3 : Test](#)

2. We will not be starting Raspberry Pi Pico coding from scratch in this user manual. There are many resources online to get you started on your journey with CircuitPython on Raspberry Pi Pico:

Adafruit Learning Guide:

[Getting Started with Raspberry Pi Pico and CircuitPython](#)





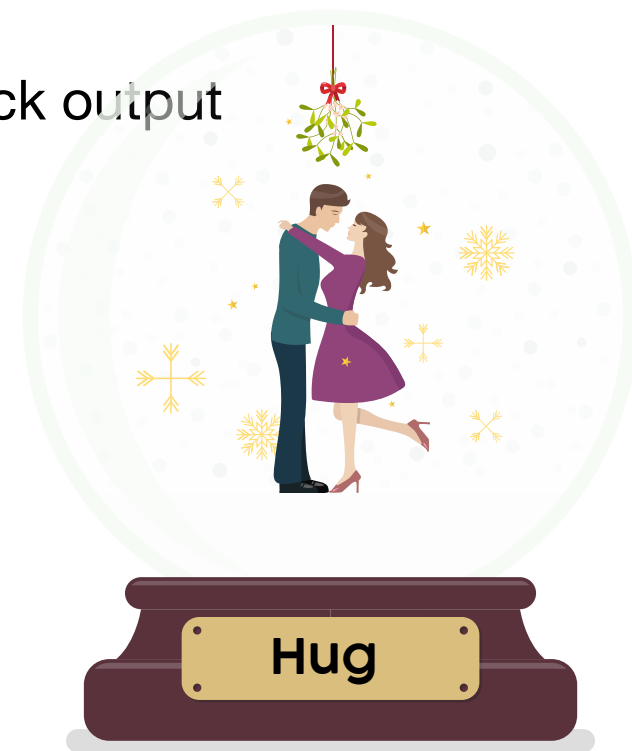
- Part 2 -

X'MAS SnowGlobe Construction



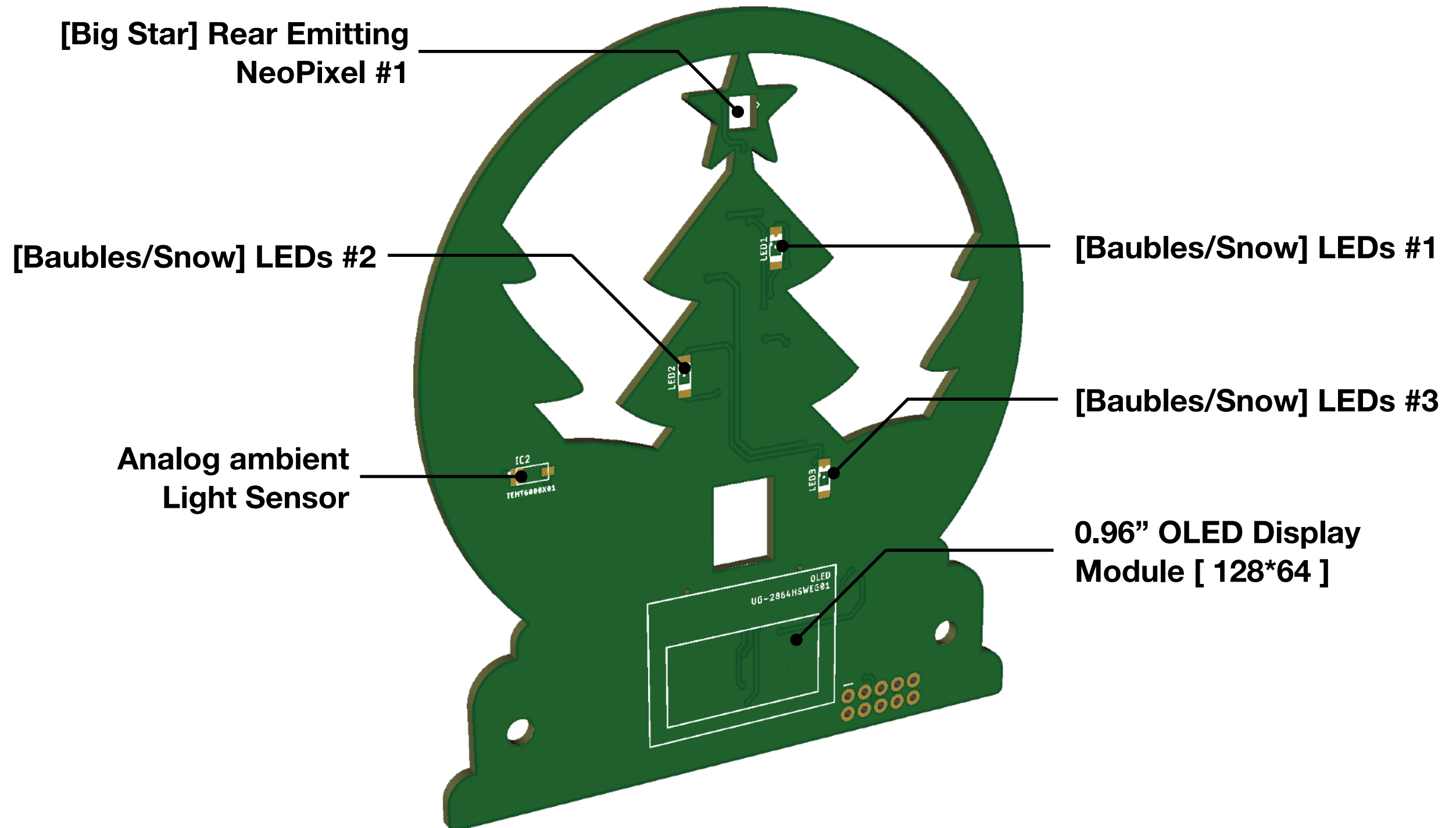
X'mas SnowGlobe Peripheral-Board

- The SnowGlobe Peripheral Board is the core of the entire build - it's where all the interaction happens.
It houses/consists of:
 - **OLED** display for messages, animations, or even a clock output
 - **NeoPixel** LEDs for colorful lighting effects
 - Standard **LEDs** for accent lighting
 - **Light sensor** for ambient-aware brightness
 - **Capacitive-touch input** for user interaction
 - 2 Pieces of Wood, a piece of Acrylic & a bunch of 3D printed trinkets
- This single board transforms your carrier board of choice, be it a micro:bit, raspberry pi Pico or the standalone WCH MCU into a fully interactive, light-up SnowGlobe centerpiece.



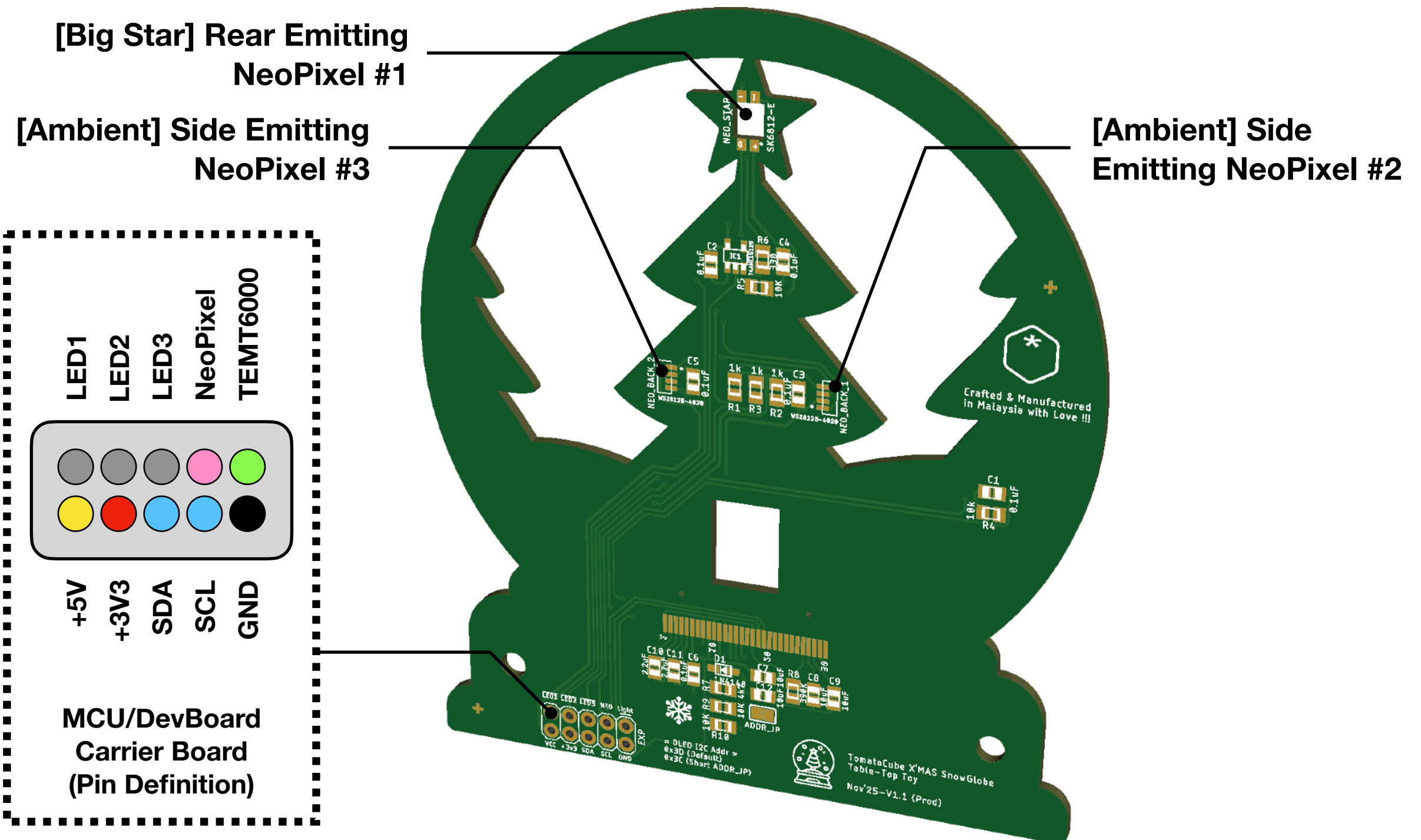
X'mas SnowGlobe Peripheral-PCB

Major Components [Front]



X'mas SnowGlobe Peripheral-PCB

Major Components [Rear]



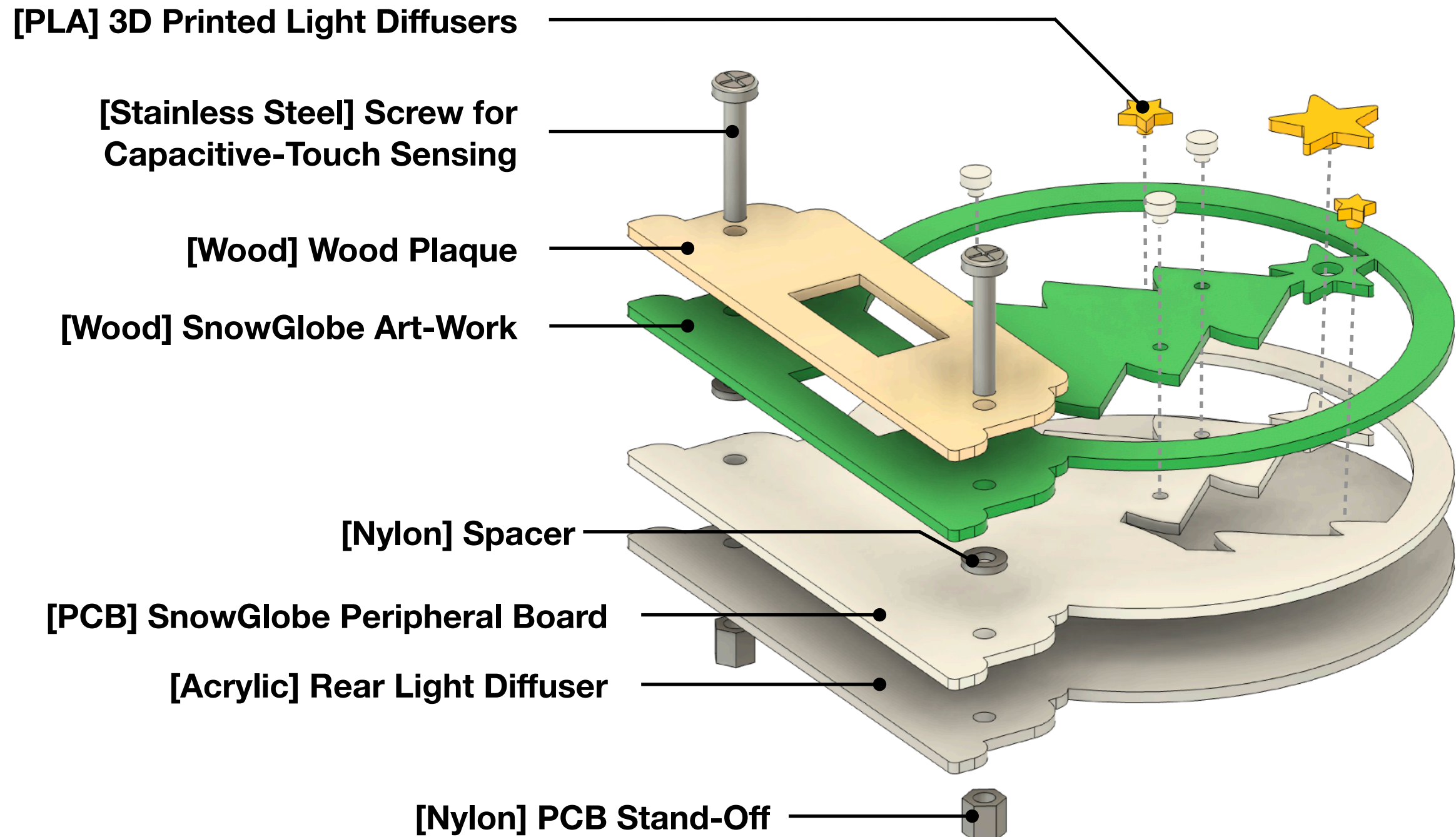
X'mas SnowGlobe

Peripheral-Board Assembly

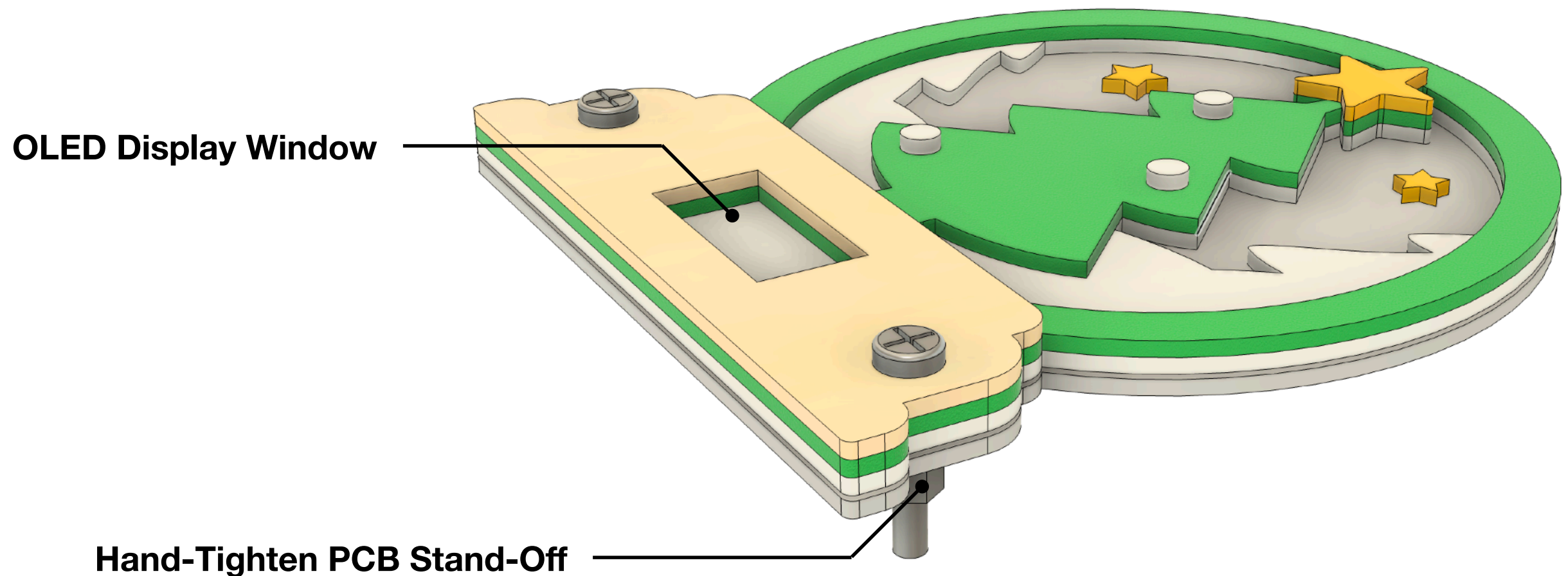
- The blown-up image/Exploded Diagram for Assembly in the next page shows how the SnowGlobe peripheral-board is arranged from top to bottom.
- **To assemble:**
 - Start from the bottom-most piece in the diagram
 - Add each layer one at a time moving upward
 - Finish by tightening the two screws and stand-offs to lock everything together
 - The diagram visually guides you, simply follow the order of layers as shown
 - Lastly, to hold everything together, we will tighten 2 screws and their respective stand-offs



X'mas SnowGlobe Peripheral-Board Components by Layer

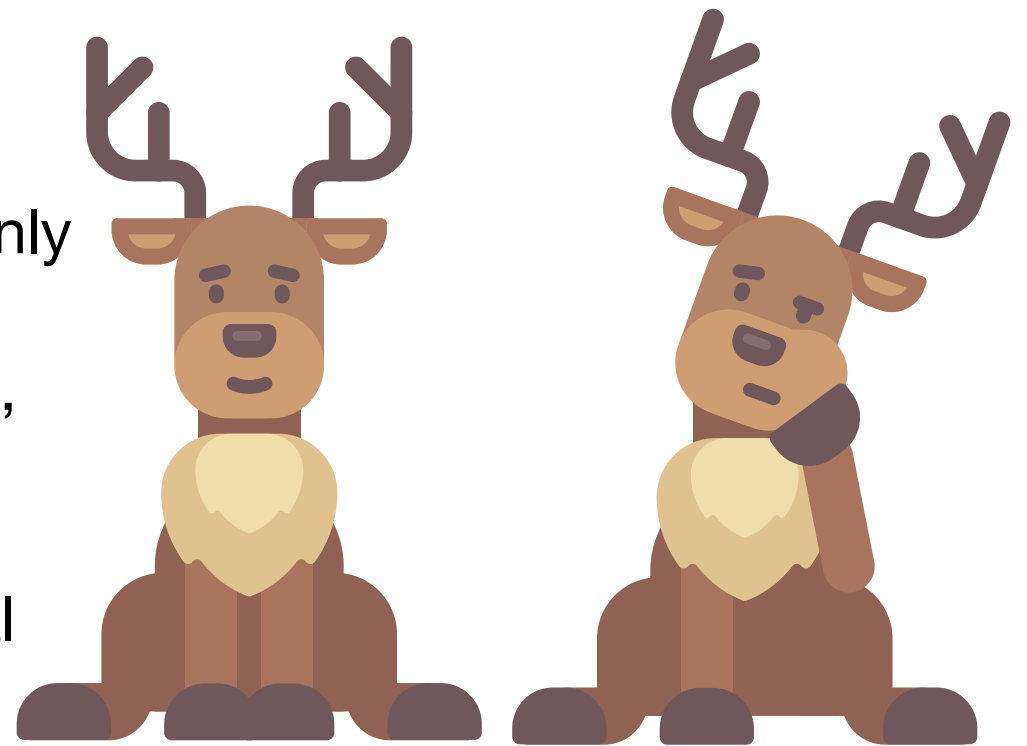


X'mas SnowGlobe Peripheral-Board Assembly Complete



Design Touches Added to Enhance the Experience

- We have painstakingly laser-cut the Wooden artwork and plaque pieces hoping to give a warm, nostalgic “real snow globe” feel
- The 3D-printed light diffusers soften LED brightness & give it a more festive glow
- Stainless-steel touch screw makes the interaction intuitive through cap-touch sensing, and it also retains the look of the SnowGlobe without intrusive Button switches
- Acrylic rear diffuser helps spread the lighting evenly
- Nylon spacers + stand-offs keep the layers clean, aligned, and sturdy
- Each layer combines electronics, craft, and visual design into one cohesive experience.

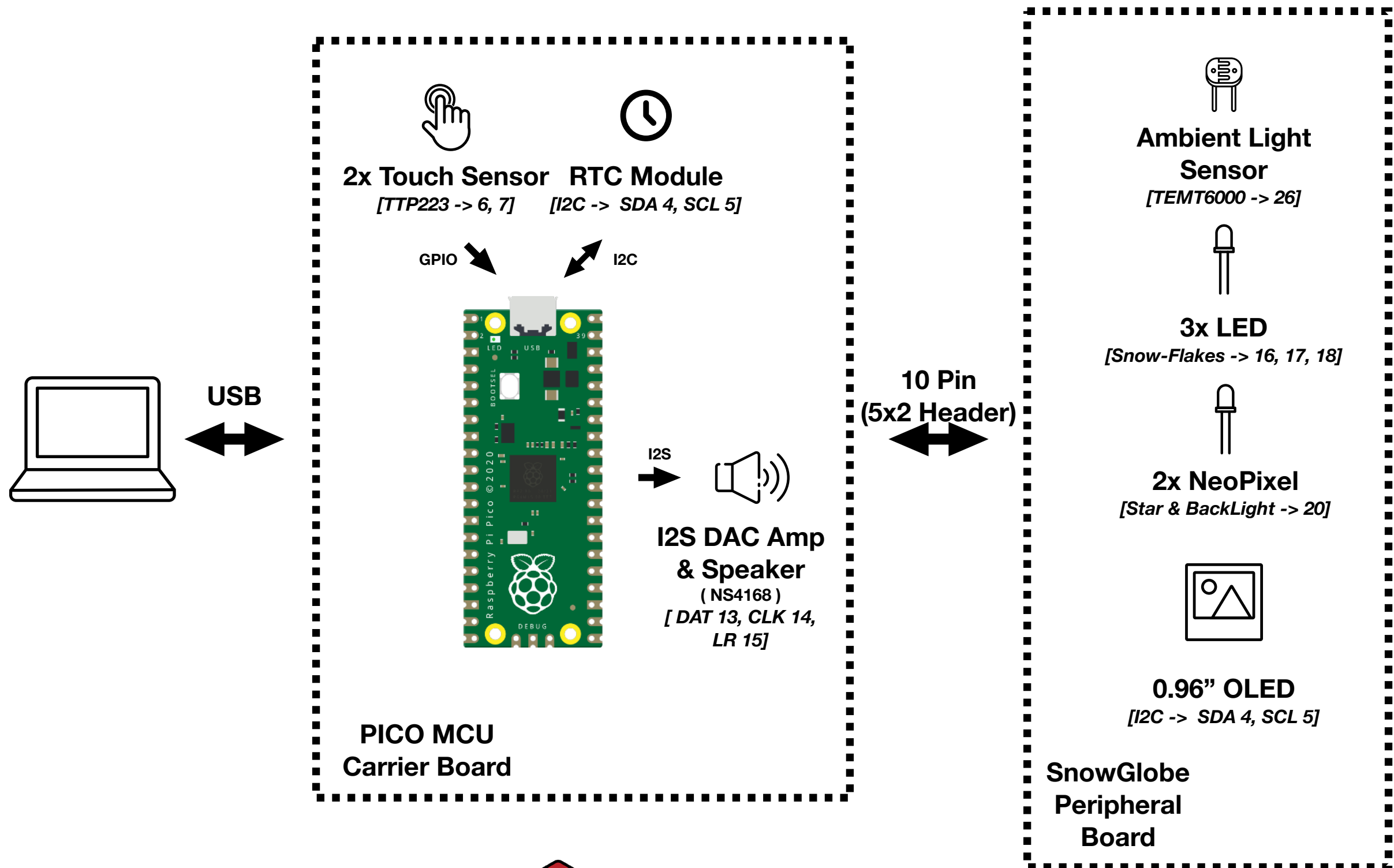


X'mas SnowGlobe Carrier Board - Raspberry Pi Pico

- The Carrier Board is the brain of the entire SnowGlobe TableTop-Toy.
 - It connects the Raspberry Pi Pico/W Board to all the interactive features of the SnowGlobe and routes power, data, and sensors through a single, easy-to-build platform.
 - Once the Raspberry Pi Pico/W Board is plugged in, this board controls every animation/display, every light, every sensor reading, and every touch.
- Key Components of the Carrier Board
 - Raspberry Pi Pico/W Board 40 Pins Pin Socket - The main socket that links to the Raspberry Pi Pico Board.
 - I2S DAC & Amplifier Chip - Decode I2S Audio Signal from the RP2040, outputting amplified analog audio signal capable of driving the onboard speaker directly.
 - Battery Power Input (Max 5V) - Battery Box input supply for cordless portable operation.
 - Power LED - Indicates the board is properly powered and functioning.
 - Touch Pads (P1 & P2) - Capacitive touch inputs connected to the 2 Screws found on the Peripheral Board
 - I²C Qwiic Expansion Socket - Plug-and-play connector for I²C sensors or peripherals



X'mas SnowGlobe Carrier Board - Raspberry Pi Pico [V1.1]



X'MAS SnowGlobe Carrier Board - Raspberry Pi Pico [Actual] [V1.1]

OnBoard Speaker

Power LED

Pico Reset Button

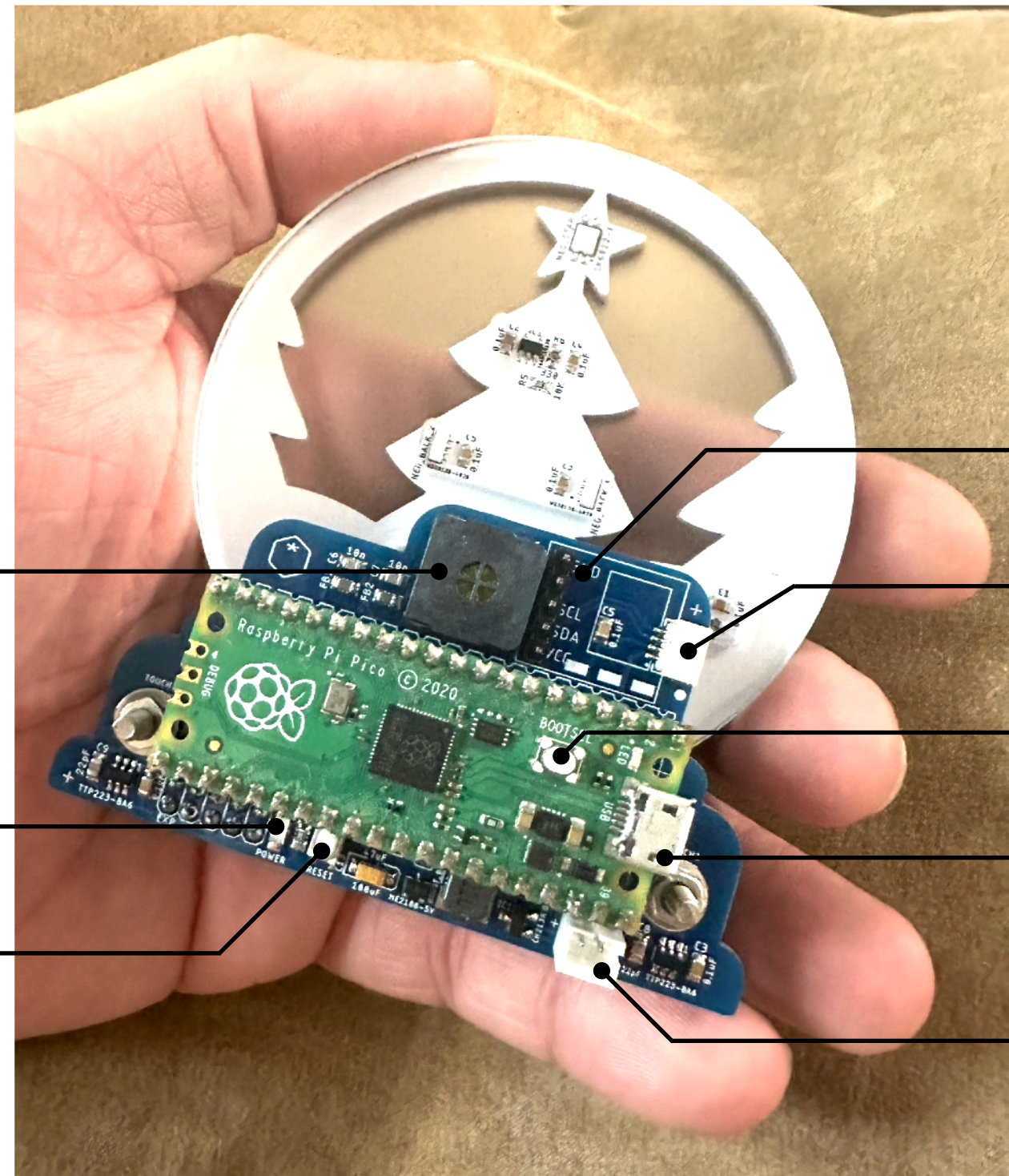
DS3231 RTC
Module PinHeader

I2C QWiic Socket

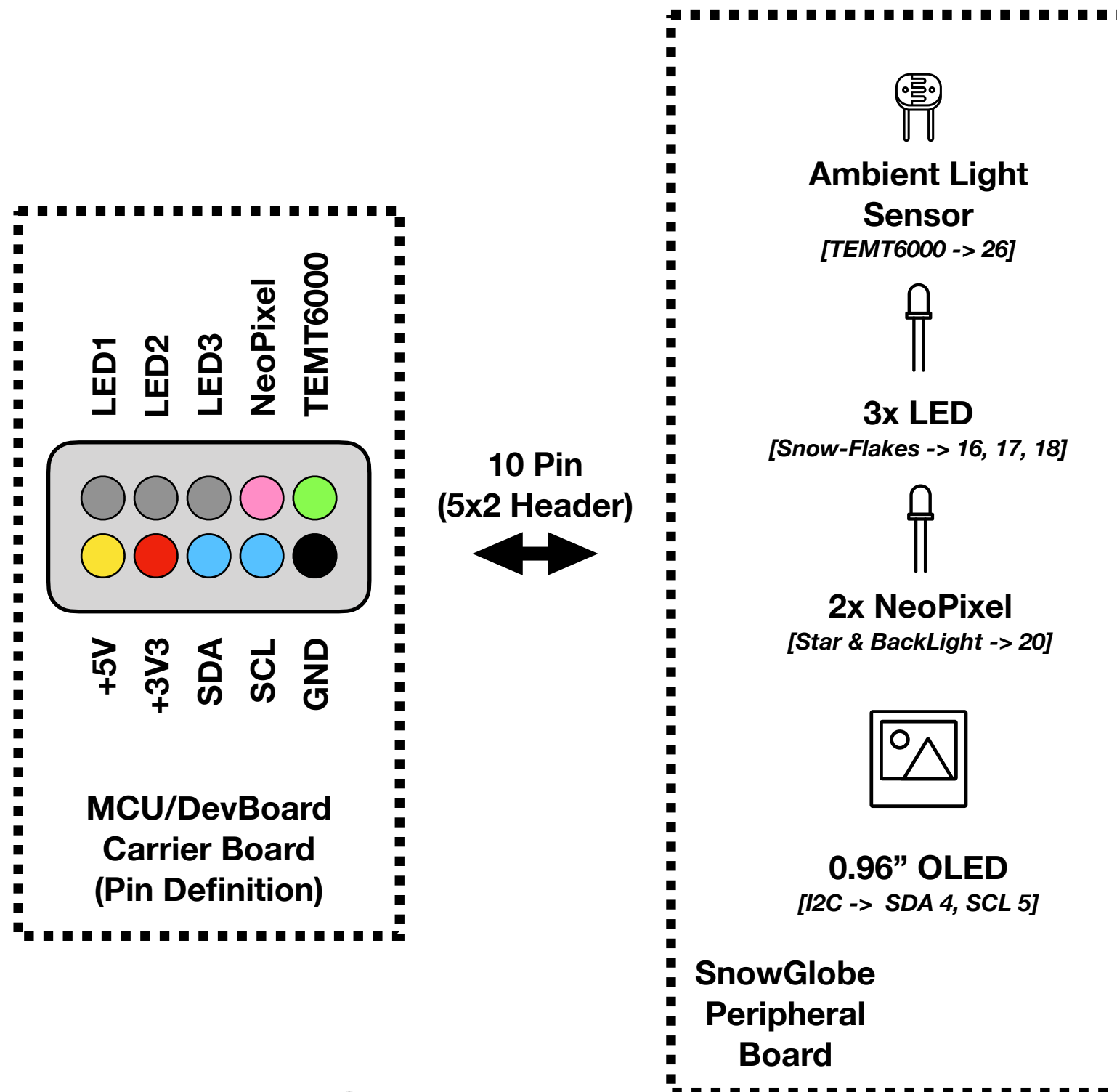
BOOTSEL Button

Pico OnBoard
microUSB Socket

Battery Power
Input (Max 5v)



X'mas SnowGlobe TableTop Toy Project Kit - SnowGlobe Carrier to Peripheral Board Pinout



X'mas SnowGlobe TableTop Toy

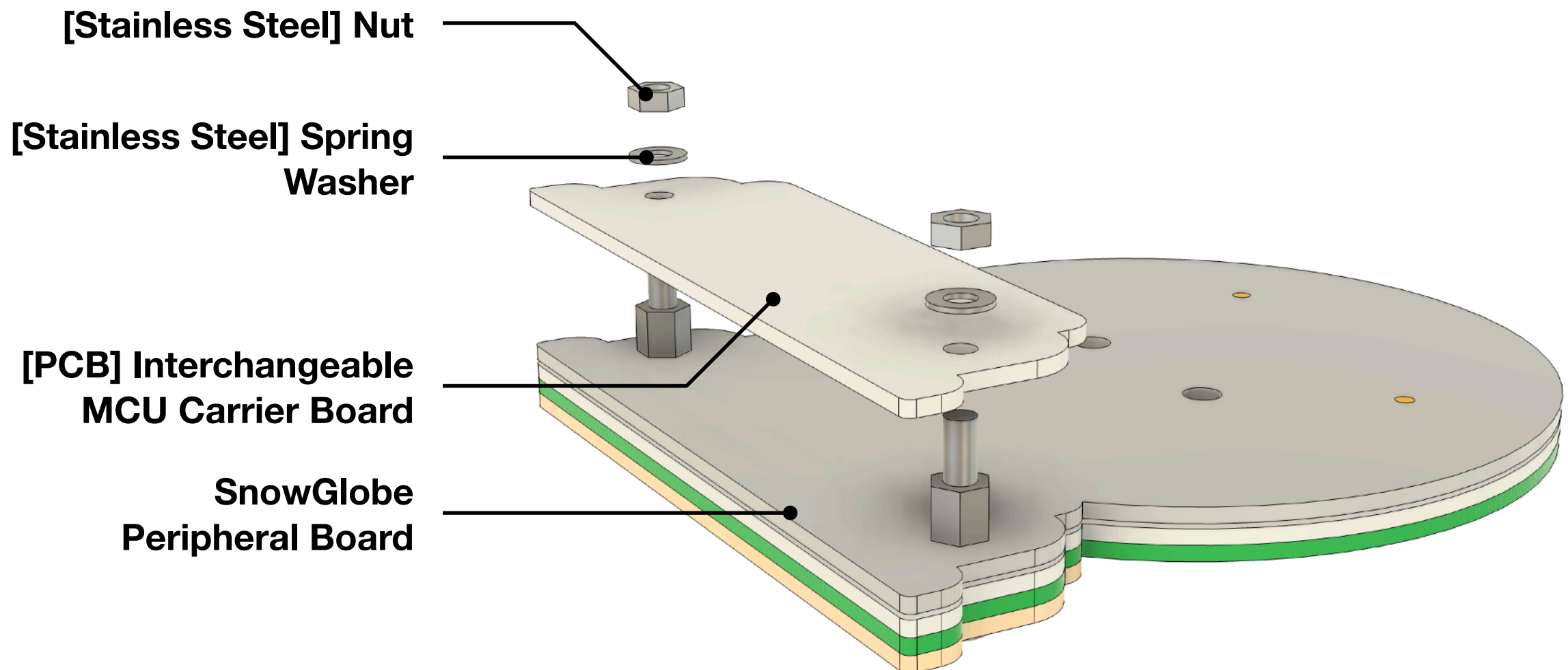
Carrier Board Assembly

- The blown-up image/Exploded Diagram for Assembly in the next page shows how the Carrier board attaches to the SnowGlobe peripheral-board.
- To assemble:
 - We will now just attach the Carrier Board to the previously assembled Peripheral Board.
 - Properly align the 5x2 pin header between the Carrier board and the Peripheral board. Firmly pushes them together.
 - Using the nut & Spring Washer provided, tighten everything together, making sure that the Spring washer provide ample contact with the Capacitive Sensing pad on the Carrier PCB.



X'mas SnowGlobe TableTop Toy

Project Kit - MCU Carrier-Board [V1.1]



X'mas SnowGlobe TableTop-Toy

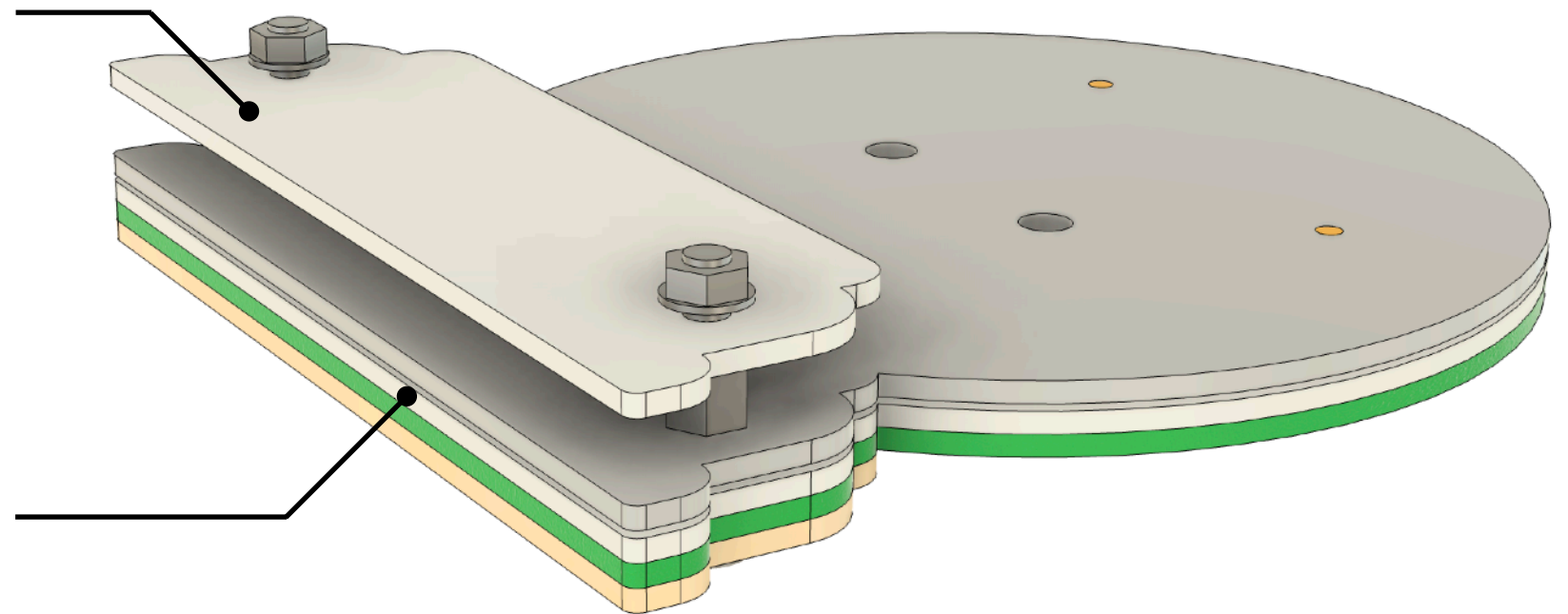
Project Kit - Fully Assembled [V1.1]

Interchangeable MCU Carrier Board

- micro:bit v2 DevBoard
- Raspberry Pico/W DevBoard
- WCH CH32x033 MCU Board

SnowGlobe Peripheral Board

- Common across multiple platform
- Reduce Cost through the use of Common Shared components



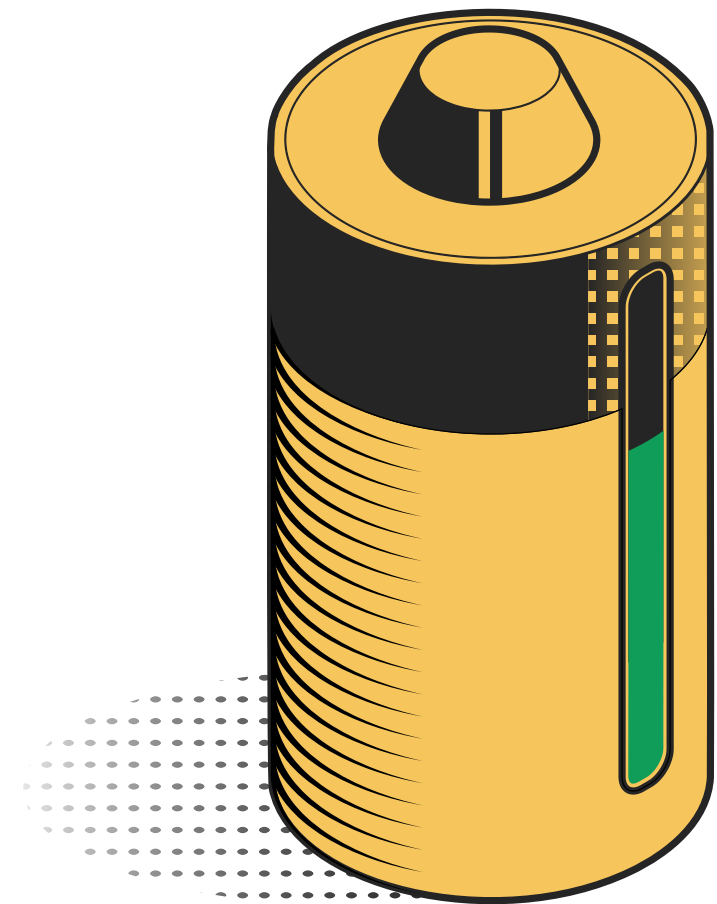
X'mas SnowGlobe TableTop Toy Project Kit - Fully Assembled [V1.1]



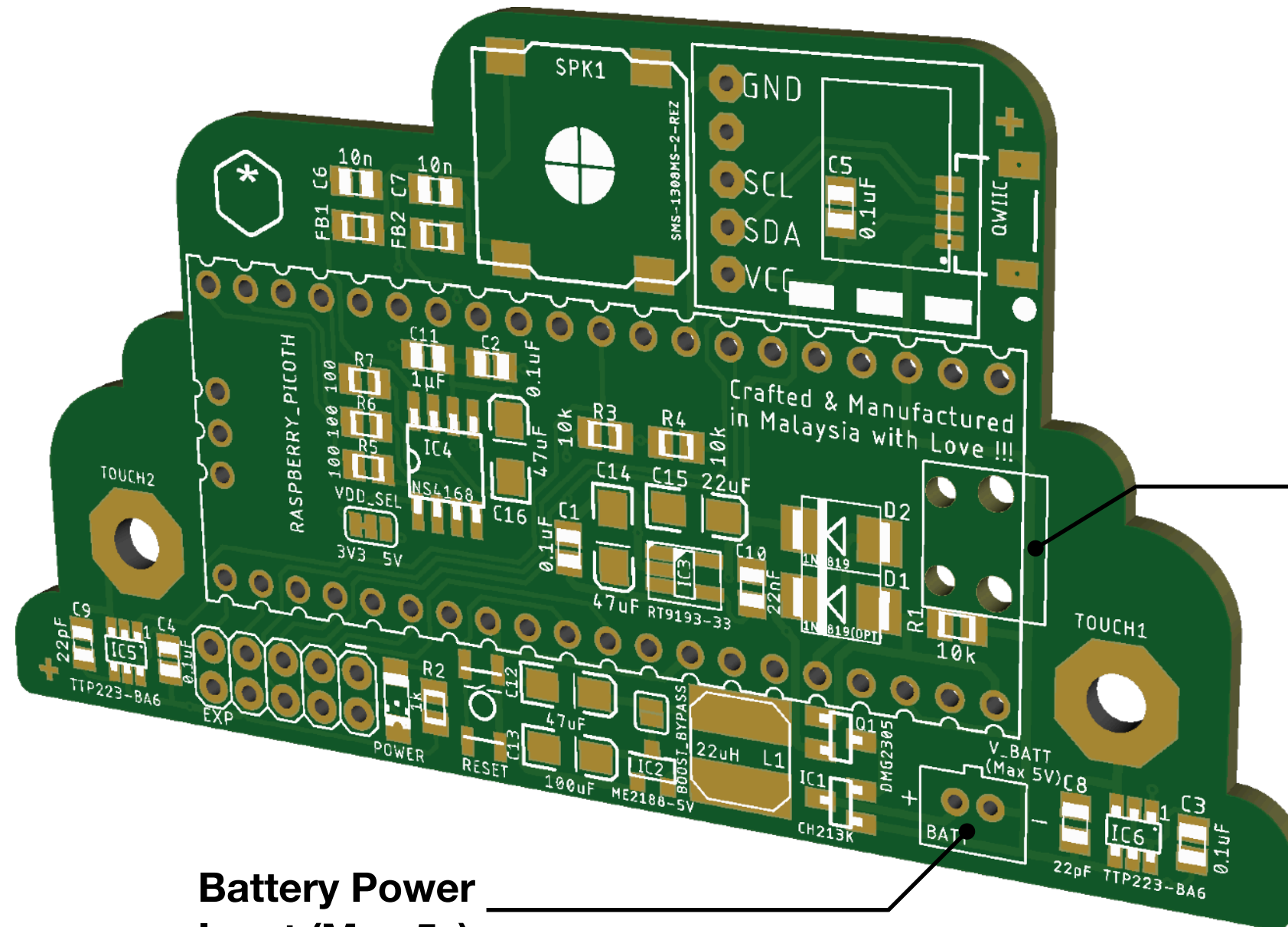
X'mas SnowGlobe

Power Option

- The SnowGlobe Carrier Board supports multiple power sources, giving flexibility for both desktop and portable use:
- USB Power (Computer or Power Bank)
 - Power the board via USB using the microUSB socket.
 - Ideal for programming, testing, classroom use, and continuous animations and X'mas melodies.
 - Provides stable 5V power to both the carrier board and the Raspberry Pi Pico Board.
- 3x AA Battery Pack
 - Connect the included AA battery box for portable, self-powered operation.
 - Great for displays, events, or locations where USB power isn't available.
 - Keeps the SnowGlobe running without cables or external devices.



X'mas SnowGlobe TableTop Toy Carrier Board - Raspberry Pi Pico - Power Options



microUSB Socket
on Raspberry Pico
Board

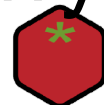
Battery Power
Input (Max 5v)





- Part 3 -

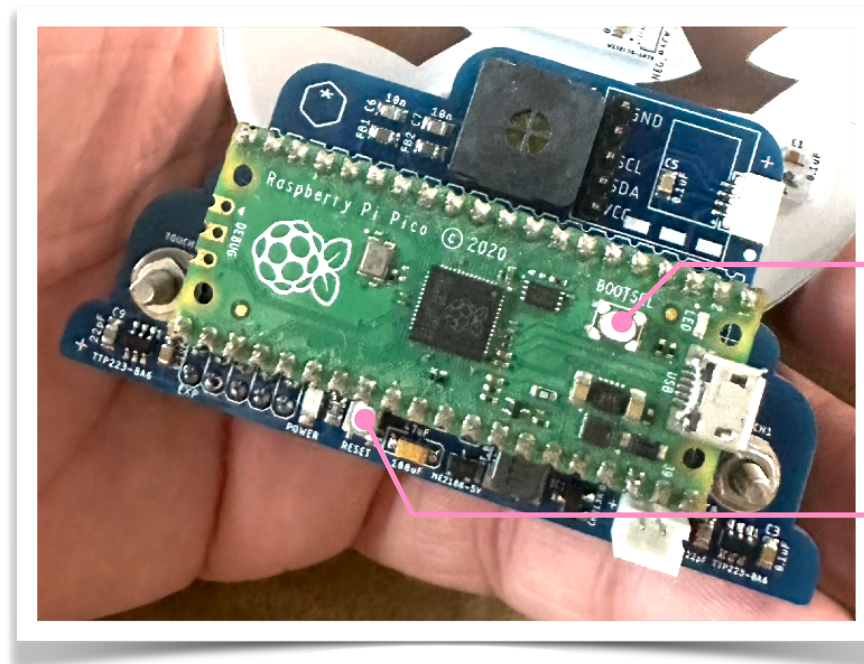
Carrier Board programming
(Raspberry Pi Pico - CircuitPython)



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Installing CircuitPython Interpreter (1/2)

- You are going to use CircuitPython on the X'mas SnowGlobe TableTop Toy. We'll use Thonny as the code editor, but CircuitPython is installed using a UF2 file.
- **Step A - Enter BOOTSEL Mode**
 - With the USB Connected to your computer.
 - Press & hold down the **BOOTSEL Button**.
 - While holding **BOOTSEL Button**, press **RESET** to reset the Pico – the **RESET button** is located on the bottom of the Carrier Board.
 - Release → **RPI-RP2** drive appears on your computer



BOOTSEL Button

Pico Reset Button



Installing CircuitPython Interpreter (2/2)

- **Step B** - Download & Install CircuitPython Interpreter
 - Download the latest CircuitPython UF2 for RP2040 from the official site Adafruit's Page - [Download for Raspberry Pi Pico / RP2040](#) or [Download for Raspberry Pi Pico2 / RP2350](#)
 - Drag and drop the corresponding **CircuitPython .uf2** file into the **RPI-RP2** drive.
 - The board will restart and now appear as a new drive, usually named **CIRCUITPY**.
- **Step C** - Use Thonny with CircuitPython
 - Open **Thonny**, then go to **Tools** → **Options...** → **Interpreter**.
 - In the **Interpreter** drop-down, choose: **CircuitPython (generic)** + correct **Serial/UART Port**
 - Create a new file & paste in the following

```
print("Hello from SnowGlobe!")
```
 - Save it to the **CIRCUITPY** drive as **code.py**.
 - press **RESET** - it will automatically run **code.py** and print the message to the serial console.



“Hello World” of embedded world - Blink

- The very first program in embedded systems is usually Blink - making an LED turn on and off. Just like “**Hello World**” in programming, Blink introduces the core ideas that every embedded project builds on:
 - Controlling hardware using code
 - Timing and delays
 - Understanding digital outputs
 - Seeing immediate physical feedback
- With the SnowGlobe and Raspberry Pi Pico Carrier Board, the **Blink** program will flash the Bottom LED Bauble/Snow - Your first step into the world of embedded electronics and interactive devices!

```
import time
import board
import digitalio

# Configure GP16 as an output pin
led1 = digitalio.DigitalInOut(board.GP16)
led1.direction = digitalio.Direction.OUTPUT

print("Blinking LED on GP16...")

while True:
    led1.value = True    # LED ON
    time.sleep(0.5)
    led1.value = False   # LED OFF
    time.sleep(0.5)
```



[GP16] LED #1

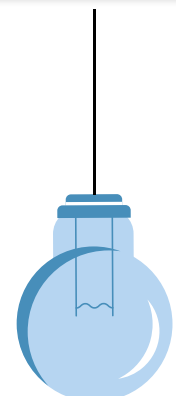


LED IS connected
to pin GP16



ON

OFF





Almost There..

- Part 4 -

CircuitPython Mini-Camp



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★ Next Step: Join the CircuitPython Self-Guided Mini-Camp!

- Ready to take your skills **further**?
You've successfully completed the basics getting started instruction in this module - now let's level up!



- 🌱 What's in the X'mas SnowGlobe TableTop Toy CircuitPython Mini-Camp?
 - Learn to **read sensors**, blink more LEDs (Including **colorful** ones), play **sounds**, and learn how to **Control Software Logic & Flow**.
 - **Bite-sized learning** designed to build real maker **confidence**.
 - Every learning topic includes **clear example**.
 - Unlocks the power💪 of **CircuitPython** step by step.
- 📍 How to get there?

Pop on over to our Github Repo for the X'MAS SnowGlobe for the Raspberry Pi Pico, look for the [Mini-camp Markdown document](#) .

